Truong Van Tinh

AI Application Engineer | Python & FastAPI

Python · FastAPI · RAG · LLM Integration · Qdrant · LangChain · Evaluation

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SUMMARY

Software engineer with 3+ years of development who applies AI at Con Cung, Vietnam's mother-and-baby retail chain (vector semantic search, n8n market-intelligence pipelines, a Claude + MCP debugging loop), and builds evaluated LLM applications in Python and FastAPI: RAG with Qdrant, LangChain, citations and groundedness checks, and LLM ticket triage with human review.

WORK EXPERIENCE

Full Stack Developer | Con Cung, Vietnam's mother-and-baby retail chain — Nov 2023 – Present

Tech: .NET, AngularJS, SQL Server, Dapper, Keycloak, n8n, Claude + MCP

- Added vector semantic search to a commerce portal so products and content are found by meaning, not exact keywords.
- Built n8n pipelines that scrape competitor data for market-intelligence reporting.
- Built a Claude + Agent Skills + MCP debug-and-test loop; used Custom GPTs and prompt engineering to speed up delivery.
- Designed retail data structures for shelves, facings and display groups that support store-layout optimization.
- Delivered AngularJS / .NET voucher workflows, 7 subscription benefit calculations and a 4-template code preview tool.
- Implemented Keycloak JWT / OAuth 2.0 / OIDC auth and role-based authorization; supported Docker and IIS deployment.
- Fixed SQL Server / Dapper retrieval defects and verified ERP API request and response mappings.

Frontend Intern | FPT — Sep 2023 – Nov 2023

- Developed Angular / JavaScript components, integrated e-commerce REST APIs and fixed mobile UI defects.

PROJECTS

Retail RAG Assistant | Personal project | Python, FastAPI, Qdrant, LangChain, Docker — github.com/tvt987/retail-rag-assistant

- Parsed Markdown, HTML, text and PDF; normalized Unicode and removed shared boilerplate before chunking with citations.
- Indexed TF-IDF sparse or SentenceTransformers dense vectors in Qdrant; answered via FastAPI and a LangChain LCEL chain.
- Abstained on low retrieval scores and fell back to cited extracts whenever the LLM failed or cited no retrieved passage.
- Evaluated 16 questions on hit rate, MRR, abstention, fact recall and groundedness, which rejects wrong numbers and uncited claims; 29 tests in two CI jobs.

Support Ticket AI Triage | Personal project | Python, FastAPI, LLM, SQLite, Docker — github.com/tvt987/support-ticket-ai-triage

- Triaged tickets (4 categories + other, priority, sentiment) with a rules baseline or an LLM whose output is schema-validated.
- Forced human review for urgent keywords, masked emails and phone numbers before the model call, token-protected the review API and stored decisions without message text.

Retail Reliability Lab | AI-assisted lab | FastAPI, Redis, SQLite, Docker Compose

- Validated Redis cache-aside (60-second TTL, invalidation, outage fallback) in 22 tests on real Redis; cut median SQLite lookup from 3.63 ms to 0.005 ms on 100,000 synthetic rows.

Other projects: Partner Integration BFF (.NET 8 idempotent partner API with RabbitMQ, Polly and 42 xUnit tests); Ecommerce-Admin (Angular 19 search-management admin with RxJS and NgRx SignalStore); Layout-Store (ASP.NET Core / PostgreSQL planogram app, contributor).

TECHNICAL SKILLS

AI applications: RAG, LLM Integration, Prompt Engineering, Document Parsing, Data Preprocessing, Chunking, Semantic Embeddings, TF-IDF, Vector Search, Document Retrieval, NLP (text retrieval), Output Validation, Abstention and Fallbacks, Human-in-the-loop Workflows, Data Pipelines, AI Agents, Retrieval Evaluation (hit rate, MRR), Fact Recall, Groundedness Evaluation

Python & tools: Python, FastAPI, LangChain, Qdrant, Vector Databases, SentenceTransformers, OpenAI-compatible APIs, SQLite, REST APIs, Input Validation, n8n, MCP, Claude, OpenAI (ChatGPT, Custom GPTs), Cursor, GitHub Copilot, OpenClaw, Agent Skills

Backend & data: C#, .NET, ASP.NET Core, Dapper, SQL Server, PostgreSQL, Redis, Database Design, Query Optimization, IIS

Security: Keycloak, JWT, OAuth 2.0, OpenID Connect, RBAC, Contact-data Masking

Testing & DevOps: Unit Testing, Integration Testing, Docker, CI/CD, GitHub Actions, Git, Logging, Monitoring/Observability, Azure, AWS

Frontend: Angular, AngularJS, TypeScript, JavaScript, RxJS, React, HTML5, CSS3, Tailwind CSS

Practices: Agile/Scrum, Code Review, Technical Documentation, Problem Solving, Cross-team Collaboration, Communication

EDUCATION AND LANGUAGES

University of Information Technology (UIT), VNU-HCM | Information Technology | Graduated 2026

English: good reading and writing of technical documents; basic spoken communication